

AI Snapshots

Warm Ups for Core Classrooms

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AI Snapshots

What are AI Snapshots?

- AI Snapshots are an assortment of classroom warm ups that will give your students a basic understanding of AI. In just 10 minutes of class time, students will learn to define, identify, and think critically about artificial intelligence.
- This set of Snapshots were based off the AI Snapshots by aiEDU, and have been modified to be used in classrooms that mainly serve Indigenous students.

Why AI?

In the next few decades, AI will affect any career field your students pursue, any consumer product they use, and policies that will shape their lives. In fact, you can bet your students are already interacting with AI in their day-to-day lives. A daily dose of AI will empower your students with the knowledge they need to succeed.

AI in Core Subjects



Real-world examples of AI are considered in the context of each of the four core subject areas. There are 2 slides for each warm up:

- Student-facing warm up slide
- Considerations for the discussion that teacher may use to prompt students. The speaker notes on this slide include links to sources and additional guidance that teachers may find helpful

Despite being organized by subject, none of the warm ups require specialized subject-matter knowledge. We welcome you to do as many warm ups as you can even if they fall outside of the subject you teach.



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Who Am I?

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Who Am I?

Use your Google search skills to find out!

Teacher and Data Scientist

- I am a data scientist for an agency that is helping the environment
- I graduated from MIT in 2018 with a Bachelors in Business Analytics
- I co-founded Lakota AI Code Camp, a summer program where students get hands-on experience building apps while incorporating Indigenous knowledge and methods



Mason Grimshaw

- Raised on the Rosebud Reservation as Sicangu Lakota
- Gave a TEDx Talk in 2023
- Passionate about preserving Indigenous languages
- Teaches at Lakota AI Code Camp



Who Am I?

Use your Google search skills to find out!

Inventor

- I am Ojibwe from the Sault Tribe
- I am the inventor of SkoBots, a personal, wearable, and interactive Indigenous language revitalization robot
- I founded The STEAM Connection, a minority and youth-led charity that has reached 800k+ kids – when I was 18 years old



Danielle Boyer

- Born in 2001
- Grew up in Michigan
- Lives in both Southern California and Michigan



Who Am I?

Use your Google search skills to find out!

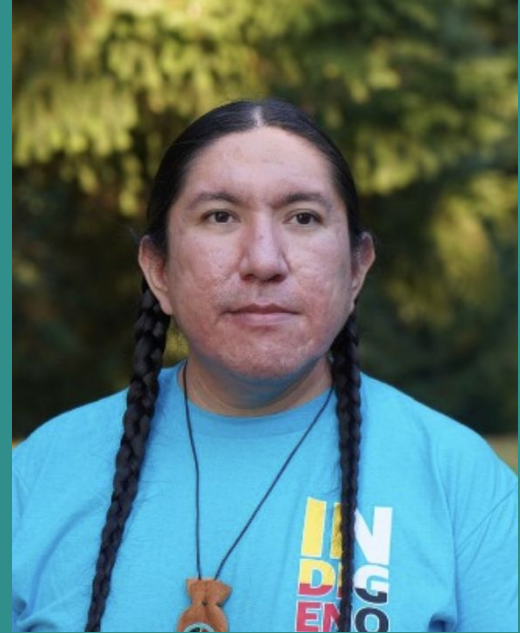
AI Poet

- I am a published poet and self-proclaimed nerd!
- I was one of the engineers who worked on Amazon's Alexa
- I am the founder of Indigenous AI, a non-profit with the goal to enable Indigenous communities to participate in Machine Learning (ML) research



Michael Running Wolf

- Raised in Montana by his grandmother who spoke Cheyenne
- Lifelong goal is to revitalize Indigenous languages using AR and VR



Fact or Fiction?



Kinship with AI

Use your Google search skills to find out: is this AI fact or fiction?

Musqueam author Quelema Sparrow uses AI to meet the character of her story. She explores whether artificial intelligence and humans can become kin. We discover the program that the AI is currently running is at odds with the human and non-human world.



Fact!



IndigiGuard

Use your Google search skills to find out: is this AI fact or fiction?

A team of Indigenous mathematicians and programmers created IndigiGuard, an AI app that blends algorithms with language preservation. Through the use of algorithms, IndigiGuard decoded the complex patterns of endangered Indigenous languages, offering translations and guaranteeing the survival of all Indigenous dialects. With each tap of a screen, IndigiGuard became not just a tool but a source of hope for Indigenous communities working to revitalize and preserve their cultural heritage.



Fiction!

Although fiction, there are **Indigenous computer scientists** actively working within their communities to blend **tradition** with **technology** and preserve their **languages**. A notable example is the **Mamutjitji Story** app, which uses a **Western Desert Dreamtime** story to teach **science**.

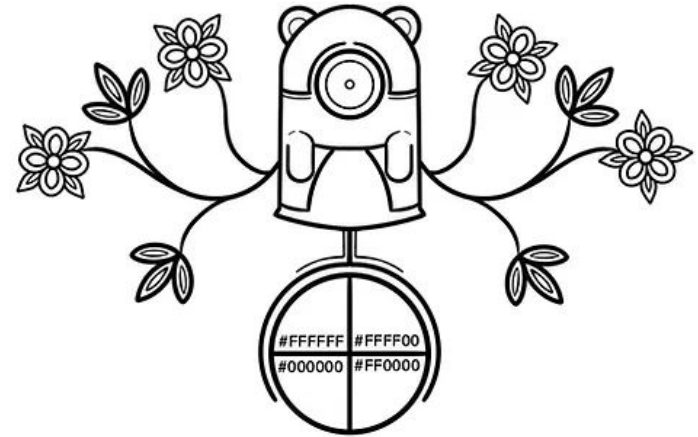
The app is available in **English** and **Ngalia**, an Aboriginal language with only three speakers. This app aims to preserve **language** and **traditional knowledge**, passing it on to the next generation.



SkoBots

Use your Google search skills to find out: is this AI fact or fiction?

SkoBots an interactive and wearable educational robot, designed to sit on your shoulder and **can learn and help teach about traditional languages**. This robot has a blend of traditional and customizable parts and responds when being approached with speech or flashing light.



Fact!

Skobots are the first cultural robot... designed entirely by Indigenous leaders.

A personalized, wearable, and interactive Indigenous language revitalization robot that senses motion and speaks our languages.



English Warm Ups

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Fake Art

In **2022**, a report found that **75% of Indigenous-style goods** were created by **non-Indigenous people**. This number will likely rise, as some people are training **AI models** on **authentic Indigenous art**, to create **AI-generated pieces**. The **artists** of the **authentic pieces** did not give their **consent** to their work being used for this purpose.



*How can **artwork** be **protected**? Who **benefits** and who is **harmed** from this **fake art**?*

Teacher Guidance

You may have considered...

- Any artist in the world is already be at risk because of the capabilities of AI. This has already come up with book covers.
- It should not be allowed to scrape for artwork or use artwork in AI models unless the artist has **given permission**.
- If non-Indigenous people are **making money** from the sales of their fake art, they are taking business away from the real, Indigenous artists -> **Furthering the colonial mindset** of taking from Indigenous communities.
- Generating fake art **threatens the cultural integrity** of the history of those artists and the communities they are from.
- Between AI's imperfections and non-Indigenous people prompting the fake art, it is likely that the **designs created wouldn't accurately reflect** the distinctive styles from specific nations and artists.

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Language Living On

Amazon Alexa operates in **9 human languages**, **Siri** in **20**. They operate in the most commonly spoken **languages** around the globe, and even provide a variety of **dialects** in **English, French**, and more. The **AI team** at **Meta** recently developed a new **AI model** that generates language based on **audio clips**.



How could you **design an AI** that helps **preserve an Indigenous language**? How could you design it so it can help preserve **any/all Indigenous languages**?

Teacher Guidance

You may have considered...

- What are the **complexities that Indigenous languages hold**, that languages like English don't?
- **Who is speaking the languages** we want to preserve and would they feel comfortable providing audio?
- Is there a framework for an AI that could be used, and **any Indigenous language** could be applied?
- App ideas for **how users would engage** with the data in an engaging way to preserve their language.
- How to **protect the data** and follow the wishes of those who provide/the communities the language is spoken in.



Hey Siri. Hey Siri. Hey Si- never mind

Many **Indigenous languages** are not recognized by mainstream **speech recognition technology**. A team of **developers** is building a new **app** addressing this by tailoring **speech recognition** to **Native American languages**.

As a **leader** within your **community**, how would you **evaluate** whether this **technology** is suitable for **widespread adoption**?

Hey Siri why are you so bad
Tap to edit >

I'm trying my best to help.
Sorry it didn't work out this time.

Teacher Guidance

You may have considered...

- How do you **measure** whether speech recognition is **effective**?
- Where will the **data** for this app come from, and how will **accuracy** be ensured?
- What are **users' experiences of the technology**: are they **enjoying** it or are they **frustrated** by it?
- How will it be built to **engage young users**, so they keep their language alive?



Beware of advice – even this

Argument A

"Developing AI to write book reviews would help people make better choices. Not only would an AI have a neutral point of view, but it would write reviews faster, too. No more waiting weeks after a book's release."

Argument B

"An AI could never appreciate a good book or recognize a bad one the way a real person could. It's important to understand the human perspective when deciding what to read next. We need real live book critics."

Which **argument** do you **agree with more**? Does it change depending on the **topic/content of the book**? **Why**?

Teacher Guidance

You may have considered...

- The AI has to **learn** to write reviews from somewhere, in this case probably old book reviews.
- What makes a **useful** book review?
- Not everyone **uses book reviews** in the same way.
- AI **might be biased** in the way it reviews books.
- Book reviewers **might lose their jobs** if the AI is very good.
- How would people feel about having **yet another algorithm** dictate the **content we interact with**?
- **Who** wrote the book, **what** is it about, and what experience does the reviewer have with the **topic and background** of the author?



Science Warm Ups

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Medicinal Plant Knowledge

Indigenous communities hold valuable knowledge about medicinal plants. Computer scientists are exploring how use AI in Indigenous communities to protect this knowledge in the digital age.



*How can AI work with Indigenous communities **preserve** their **traditional knowledge of medicinal plants** while making sure it is **respected** and not misused?*

Teacher Guidance

You may have considered...

- Discuss the importance of **cultural preservation** and the ethical considerations when integrating AI into traditional practices.
- Explore ways in which Indigenous communities can actively participate in the development and use of AI technologies to **safeguard their knowledge**.
- Consider the role of education and collaboration in ensuring that AI initiatives **respect and benefit Indigenous cultures**.

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Computer Science

Human–Nonhuman Connections with SkoBots

SkoBots are AI language bots designed to have personalities, but how comfortable would individuals be interacting with them? Especially in Indigenous communities where **cultural connections** are significant.

*How do you feel about using a language bot like SkoBots that **acts like a person**, and what would make you feel **comfortable** or **uncomfortable** when using it?*



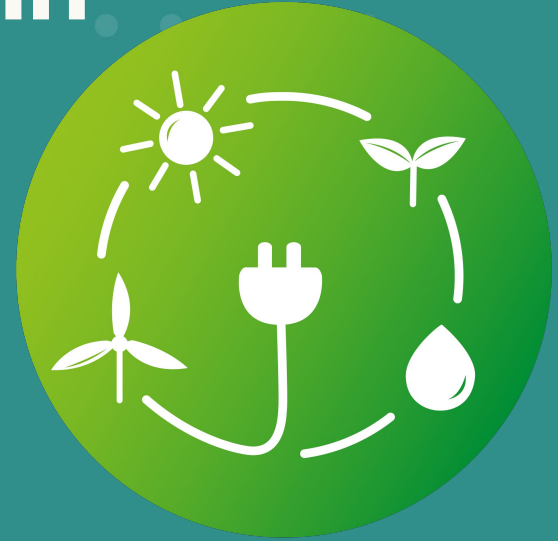
Teacher Guidance

You may have considered...

- How this technology could impact **human relationships and communication.**
- The importance of **respecting cultural beliefs** and values when using AI technology like SkoBots (designed by Native youth for Native youth!)
- Weighing the potential benefits with the potential drawbacks and thinking about **what guides us in forming opinions.**

AI-Driven Sustainability in Your Community

Indigenous communities often face unique challenges in accessing **sustainable energy sources**. AI technology can play an important role in identifying the most suitable renewable energy solutions based on **local conditions**, such as wind patterns and sunlight availability.



*How can AI be used to **assess** the most effective **sustainable energy sources** for your community, considering factors like wind patterns and sunlight?*

Teacher Guidance

You may have considered...

- Why it's important for everyone in the community to help decide on ways to use clean energy and how **AI can help us all work together** to make our future more sustainable.
- How AI can help decide **where to put things** like wind turbines to make them work best in different parts of your area.
- The importance of **involving community members** in decision-making processes regarding sustainable energy initiatives.
- What, if any, are the unintended **implications** of each of these energy sources?



Connections to Star Stories

Many Indigenous cultures have rich oral traditions and stories that describe the origins and importance of stars, planets, and constellations. These stories offer a **unique lens in the scientific understanding of the universe**. With the advancement of AI, this could help to **preserve, share, and connect with these cultural narratives**.



*How would you **design** an AI that can **share** stories and **knowledge about the stars**?*

Teacher Guidance

You may have considered...

- Discussing the importance of **preserving and sharing these cultural stories**, and how AI could possibly play a role.
- **Potential capabilities of AI** such as **storytelling** through text, audio, visuals...etc, **interactive exploration**, and connections to relevant **scientific information and data**.
- Ways to **make sure the system is respectful, accessible, and engaging** for diverse audiences.
- Data privacy, cultural sensitivity, and appropriate use of traditional knowledge.



Math Warm Ups

Navajo Code Talkers

Navajo Code Talkers were key to the **allied victory** of **World War II**. They were able to **encrypt** and **decrypt** messages in **1/10 of the time** computers were able to at that time. Even though the **Navajo language** is complex and wasn't written at the time, the **code breakers** added another layer with **word substitution**. The "**dictionary**" used to decipher a code is now available.



*In what ways do you see **Indigenous youth** shaping the **future of technology and AI**, both in their **communities and worldwide**?*

Teacher Guidance

You may have considered...

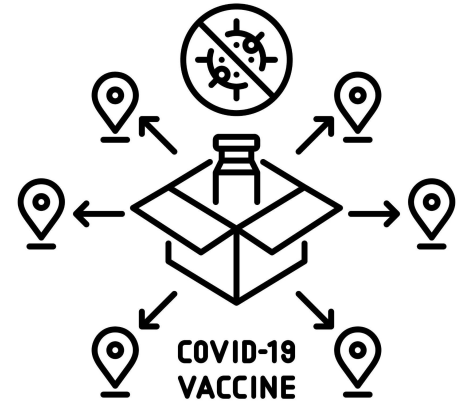
- The people who create **tech** should represent those who **use** it.
- **Indigenous communities** have unique **histories, values, and needs** that “**big tech**” isn’t always considering.
- **Young Indigenous people** can build solutions unique to their **communities**.
- **Young Indigenous people** can join **tech teams** and open up **representation** in those spaces.



First served

COVID-19 illustrated the complications with **large-scale vaccine distribution**. To decide which communities to prioritize, officials consider **hundreds of statistics** like population density, residents' age, and number of hospitals. The process is **time-consuming** and statistics are **often inaccurate**. The focus was largely on the world's most populated areas.

How would you **design an AI** that can learn to accurately get the statistics that matter to **your community**?



Teacher Guidance

You may have considered...

- How to determine **which factors make someone high-risk**
- **Demand** for the vaccine itself.
- **Common comorbidities** associated with COVID.
- Potential **bias in the statistics** used to make decisions.
- **Deadly impacts** of getting things wrong and **liability** around them.



Starting early

Imagine you're starting **Math+AI4**, a non-profit that aims to **engage middle school students** in your community in the **mathematics of AI**. You're having trouble recruiting students. You've found that students, parents, and teachers are unaware of the vast impacts AI will have on their lives, so they don't seek out your programs.

How would you **design an AI** that can learn to solve this problem using **survey data** about students' interests?



Teacher Guidance

You may have considered...

- **Why** there is a lack of interest?
- What might **engage prospective students**.
- How to **collect the data** you need from minors.
- How to **include perspectives** of teachers and families.
- How to **measure success**.

The Real Salvator Mundi

Can math help us figure out **if a painting is authentic**? Researchers divided up Da Vinci paintings into **350x350 pixel squares** and trained an AI to **recognize his handiwork**.



*How could AI help **preserve the integrity of Indigenous art**, and why is it important to do so?*

Teacher Guidance

You may have considered...

- An AI could be trained on **markers of authenticity** based on tribe, artist, and more.
- **Community Input.** If AIs are trained on Indigenous art, that data is “out there”. Do members of the communities want to put more out there in hopes this AI works, or do they prefer not to?
 - Not everyone in a community will agree; **who** makes the decision?
- Figuring out if an image has been **digitally altered**.
- **Authorship is complicated**, with frequent collaboration and ease of copying information.



Social Studies Warm Ups

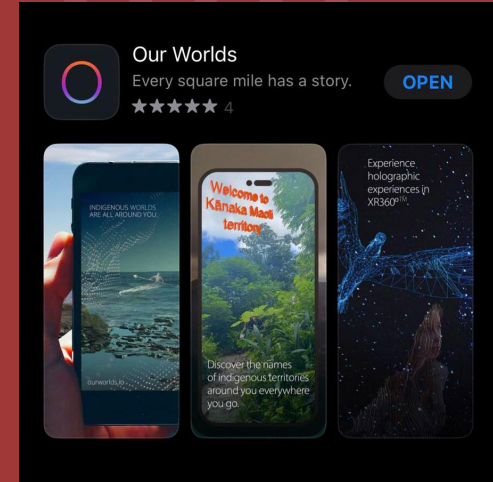
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AI for Equal Access to Museums

OurWorlds, an extended reality platform showcasing historical and cultural experiences, aims to make **educational content accessible globally**. Disparities in access to museum exhibits between **major metropolitan areas** and **smaller cities/rural areas** highlight the need for innovative solutions like AI to bridge this gap.

*How would you design an AI system that uses **images of museum exhibits** to create a virtual experience **accessible** to individuals in underserved areas, **including Indigenous communities**?*



Teacher Guidance

You may have considered...

- The **effects** of **AI** and **extended reality technologies** can help make **Indigenous cultural knowledge** and **artifacts** more accessible.
- AI can **analyze and interpret images** of Indigenous cultural exhibits to generate interactive virtual experiences that authentically represent the traditions, stories, and contexts of these artifacts.
- **Involving Indigenous communities in the design** and development of AI-powered virtual experiences to support the preservation of cultural integrity and knowledge.

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Age Bias in Healthcare AI

Recent research indicates that healthcare **AI may not properly support elders**. This issue of **age bias** in healthcare AI raises concerns about **equitable access to technology-driven healthcare** solutions for all age groups, including Indigenous communities.



*How can we **address age bias in healthcare AI** to **support our elders** in receiving the necessary support and benefits from tech advancements in healthcare?*

Teacher Guidance

You may have considered...

- Indigenous communities may have **unique cultural practices and languages** that aren't always considered in healthcare AI.
- **Access to technology** might be limited in some communities, making it harder for older individuals to use healthcare AI.
- Think about the **specific needs** and cultural sensitivities
- Making sure the technology is **respectful of traditions and languages**, and easy for older members of these communities to use.
- **Involve community members** in the development process to make sure it meets the needs of our elders
- By including the community's **input and feedback**, we can create **technology that respects their culture** and addresses any biases that might exist.

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Access to Healthy Food Options

In many communities, **access to affordable and nutritious food** is limited. This issue worsens health disparities and challenges residents' ability to maintain a **balanced diet**.



*How can AI technology be used to improve **access to healthy and affordable food**, and **what role can community members play** in this process?*

Teacher Guidance

You may have considered...

- Why it's essential for **Indigenous community members to help decide on ways to get better food access**, making sure it fits their culture and sustainability.
- Using AI as a tool to **analyze** food disparities and which communities might need more access to healthier food options.

Preserving artifacts

Indigenous cultures around the world have diverse collection of **artifacts** that hold **deep significance** and **provide insights into the histories of these peoples**. Unfortunately, many of these artifacts have been damaged, lost, or removed from their original context over time. AI systems could be used to help **preserve**, **restore**, and **connect people** with these important artifacts.

*How could AI technology be adapted to help **preserve**, **document**, and **share artifacts** and their **cultural significance**?*



Teacher Guidance

You may have considered...

- Consider **ethics** in terms of data privacy, cultural sensitivity, and traditional knowledge.
- Brainstorm **potential capabilities of AI systems**:
 - Automated restoration
 - Identification of and location of artifacts
 - Multimedia storytelling
 - Connections to relevant historical data
- Having **people connect with these artifacts** can foster a greater understanding and appreciation.



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